Summer Math Adventure

W3
D1

Week 3 · Day 1 | Multi-Digit Multiplication: Mastery

Name: ____________________ Date: ____________________

MULTIPLICATION

Standard Algorithm – 2-digit × 2-digit:
47 × 38: Multiply 47 × 8 = 376, then 47 × 30 = 1,410. Add: 376 + 1,410 = 1,786.
Shortcut check: Estimate first! 47 × 38 ≈ 50 × 40 = 2,000. Answer 1,786 is close. ✓
Area model alternative: Split 47 = 40 + 7, 38 = 30 + 8. Fill in the 4 boxes and add.

Part A – Standard Algorithm

1. 73 × 46 =

2. 258 × 37 =

3. 614 × 83 =

4. 307 × 59 =

5. 4,025 × 14 =

6. 528 × 406 =

Part B – Area Model

Fill in the area model, then add the parts.

7. 63 × 47 using the area model:

	40	7
60		
3		

Total: ___ + ___ + ___ + ___ =

____

Part C – Word Problems (Multi-step)

8. A theater has 48 rows of seats. Each row has 36 seats. Tickets cost $12 each. If the theater is completely full, how much money is collected?

Answer: $

Summer Math Adventure

W3 D2

Week 3 · Day 2 | Long Division: 4-Digit ÷ 2-Digit

Name: ____________________ Date: ____________________

LONG DIVISION

Long Division Steps — remember "DMSB":
Divide → **M**ultiply → **S**ubtract → **B**ring down (then repeat)
Example: 1,344 ÷ 16 → Ask: how many 16s in 13? → 0. How many 16s in 134? → 8 (8×16=128). Subtract → 6. Bring down 4 → 64. 64÷16=4. Answer: 84.
Always check: quotient × divisor + remainder = dividend.

Part A – Divide (show each step)

1. 5,376 ÷ 24 =

Check: ____________________

2. 7,056 ÷ 48 =

Check: ____________________

3. 4,891 ÷ 13 =

Q: ____________________

R: ____________________

4. 9,072 ÷ 36 =

Check: ____________________

Part B – Estimate the Quotient First

Round the divisor to a friendly number, estimate, then calculate exactly.

5. 8,463 ÷ 27 Estimate: ____________________

Exact: ____________________

Summer Math Adventure

W3
D3

Week 3 · Day 3 | Factors, Multiples, Prime & Composite Numbers

Name: ____________________ Date: ____________________

FACTORS & MULTIPLES

Factor: a number that divides evenly into another. Factors of 12: 1, 2, 3, 4, 6, 12.
Multiple: the result of multiplying by a whole number. Multiples of 6: 6, 12, 18, 24...
Prime: exactly 2 factors (1 and itself). Examples: 2, 3, 5, 7, 11, 13...
Composite: more than 2 factors. Example: 12 has 6 factors → composite.
GCF: Greatest Common Factor. **LCM:** Least Common Multiple.

Part A – List All Factors

1. Factors of 36: ____________________

2. Factors of 48: ____________________

3. Factors of 60: ____________________

4. Factors of 100: ____________________

Part B – Prime or Composite?

Write P (prime) or C (composite). For composite numbers, list one factor pair (other than 1 × itself).

Number	P or C?	Factor pair (if composite)
17		
24		
37		
49		
51		
97		

Part C – GCF and LCM

Use factor lists or prime factorization (factor trees).

5. GCF of 24 and 36 = ☐

Factor trees or lists:

6. GCF of 45 and 60 = ☐

7. LCM of 4 and 6 = ☐

8. LCM of 8 and 12 = ☐

Summer Math Adventure

W3 D4

Week 3 · Day 4 | Order of Operations (PEMDAS)

Name: ______________________ Date: ______________________

ORDER OF OPERATIONS

PEMDAS: Parentheses → **E**xponents → **M**ultiply & **D**ivide (left to right) → **A**dd & **S**ubtract (left to right)

Example: $3 + 4 \times (2 + 1)^2 - 5$

$= 3 + 4 \times 3^2 - 5$ (parentheses first)

$= 3 + 4 \times 9 - 5$ (exponents)

$= 3 + 36 - 5$ (multiply)

$= 34$ (left to right add/subtract)

Part A – Evaluate Step by Step

Show EACH step. Circle the operation you do first.

1. $5 + 3 \times 4 =$

Step 1: ______________________

Step 2: ______________________

2. $(5 + 3) \times 4 =$

Step 1: ______________________

Step 2: ______________________

3. $20 - 12 \div 4 + 2 =$

4. $(20 - 12) \div 4 + 2 =$

5. $3^2 + 4 \times (6 - 2) =$

6. $48 \div (2 \times 3) + 5^2 =$

Part B – Add Parentheses to Make It True

Place parentheses in each expression to make the equation correct. You may need to try a few spots.

7. $3 + 5 \times 2 - 1 = 15$

8. $10 - 2 \times 3 + 1 = 25$

9. $24 \div 4 + 2 \times 3 = 12$

10. $5 \times 3 + 4 - 2 = 33$

Summer Math Adventure

W3 D5

Week 3 · Day 5 | Multiplication & Division Word Problems + Full Review

Name: ______________________ Date: ______________________

OPERATIONS REVIEW

Word problem strategy: Read → Identify what's asked → Choose operation → Estimate → Solve → Check if answer makes sense.

Part A – Mixed Computation Sprint

1. $784 \times 56 =$

2. $9{,}408 \div 28 =$

3. $5 \times 3^2 - (4 + 2) =$

4. GCF(18, 30) = ______________________

5. LCM(6, 9) = ______________________

6. Is 91 prime? Circle: YES / NO
Factor pair if composite: ______________________

Part B – Multi-Step Word Problems

7. A pool holds 35,280 gallons. A pump fills it at 840 gallons per hour. How many hours will it take to fill the pool completely?

Answer: ______________________ hours

8. Marcus earns $14 per hour. He works 38 hours one week and 42 hours the next. He spends $350 on rent. How much does he have left after paying rent over both weeks?

Answer: $ ______________________

9. A store has 2,400 apples, 1,800 oranges, and 3,000 bananas. They want to pack fruit bags where every bag has an equal number of each type of fruit. What is the greatest number of bags they can make? (Hint: use GCF)

Week 3 Quiz – Multiplication, Division & Number Theory

W3 QUIZ

Topics: Multi-digit ×÷ · Factors/Multiples · GCF/LCM · Order of Operations

Name: ______________________ Date: ______________________

Score: _______ / 20

Section 1 – Multiplication & Division (8 pts)

1. 437 × 68 =

2. 6,720 ÷ 32 =

3. 5,036 ÷ 14 =   Q: ___ R: ___

4. $4 \times (3 + 7)^2 \div 5 =$

Section 2 – Factors, Primes, GCF, LCM (6 pts)

5. List all factors of 42:

____

6. Is 83 prime or composite?

____

7. GCF(32, 48) =

____

8. LCM(9, 12) =

____

9. **(2 pts)** You have 24 red beads and 36 blue beads. You want to make identical bracelets using all beads (each bracelet has the same mix). What is the maximum number of bracelets? How many of each color per bracelet?

Bracelets:

____

Red each:

____

Blue each:

____

Section 3 – Word Problems (6 pts)

10. **(3 pts)** A farmer plants 45 rows of corn with 68 plants in each row. He harvests 2,856 plants and stores them in bags of 24 plants each. How many full bags does he fill?